Mahima Sundararajan

4134598891 mahimasundarp@gmail.com mahimasundarp mahima1911 Portfolio

Education

University of Massachusetts at Amherst Amherst, MA
MS in Computer Science *Aug. 2023 – May 2025*

Cumulative GPA: 3.95/4

Relevant Coursework: Machine Learning, Secure Distributed Systems, Systems for Data Science, Software Engineering, Statistics, Database Design and Implementation

Anna University Chennai, TN
B-Tech Information Technology *Aug. 2018 – Jun 2022*

Cumulative GPA: 8.9/10 (3.68/4)

Relevant Coursework: Database Management Systems, Advanced Data Structures, Operating Systems, Cloud Computing

Activites: Member of Google DSC, IEEE, Google Women Techmakers, Google Developers Group, Coding Club, ML Tech Club;

Work Experience

Associate Software Engineer Jan 2022 – Aug 2023
Kaleris *Chennai, TN*

- Applied **Agile methodologies** in the pivotal introduction of multi-tenancy, enhancing our product's capabilities.
- Maximized operational efficiency by implementing dynamic real-time Power BI dashboards for 25 international terminals, revolutionizing performance insights.
- Remediated bugs, honed processing, and boosted efficiency via advanced **ETL** enhancements using **Pentaho** and **SQL**.
- Redesigned data processing workflow by implementing **Java** and **Kafka** solutions.
- Ensured flawless version migration, while propelling product enhancement through meticulous **Groovy-based** testing.

Software Development Analyst Intern May 2021 – Aug 2021
LatentView Analytics *Chennai, TN*

- Executed end-to-end development of a **VBA-powered Excel** application, serving a global user base of 250,000, involving requirement analysis, design, and coding from scratch.
- Designed and deployed an automated multi-tenant chatbot using **ReactJS**, concurrently fine-tuning communication efficiency via a **Python-Flask RestAPI** backend.

Projects /Publications

Analysis of Lombard effect on Speech Emotions using Machine Learning

- Orchestrated and executed experiments examining the impact of external noise on human emotions, using a curated dataset of *30 individuals* spanning diverse age groups. Implemented more than **10 advanced Machine Learning techniques** and a **Convolutional Neural Network** in Python to increase overall accuracy to 90%.
- *Published* in an international Scopus-indexed journal, detailing how background noise affects vocal emotions.

SEEDB

- Spearheaded the development of SEEDB, a **robust database system** designed to manage complex data efficiently. Employed **advanced SQ**L techniques for optimized data retrieval and **Python** for backend processing, enhancing data handling capabilities. The system supports detailed data analysis, ensuring precise information management and retrieval for user-specific queries.

Code Summarization and Evaluation

- Developed a code summarization system using **GPT-3.5 Turbo**; integrated user authentication, streamlined code input, and created a feedback mechanism to assess summary quality, enhancing overall code understanding.

CricInsight 2023

- Utilized **Python** and **PySpark** to analyze player data (batting, bowling, historical) and create KPIs for predicting Cricket World Cup match winners. Employed **6 ML models** for accurate outcome forecasts.

Technical Skills

Languages: C++, Python, Java, C, HTML, CSS, JavaScript, SQL, R
Specializations: Data Analysis, Machine Learning, Mobile-App development, Web Development, Generative AI
Database Management Systems: Oracle SQL, MySQL, Excel, Firebase, Apache Spark, Microsoft Azure, MongoDB, Hadoop
Libraries & Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, Keras, Git
Data Visualization Tools: Power BI, Tableau, Plotly,

Achievements

First Class with Distinction – SSNCE, 2022
National Finalist (6th) – Wells Fargo ML Hackathon, Shaastra'21 IIT M, 2021
Winner (Placed 5th in Department) – Merit Scholarship, SSNCE, 2020 - 2021
Machine Learning Scholarship – AWS ,2021